

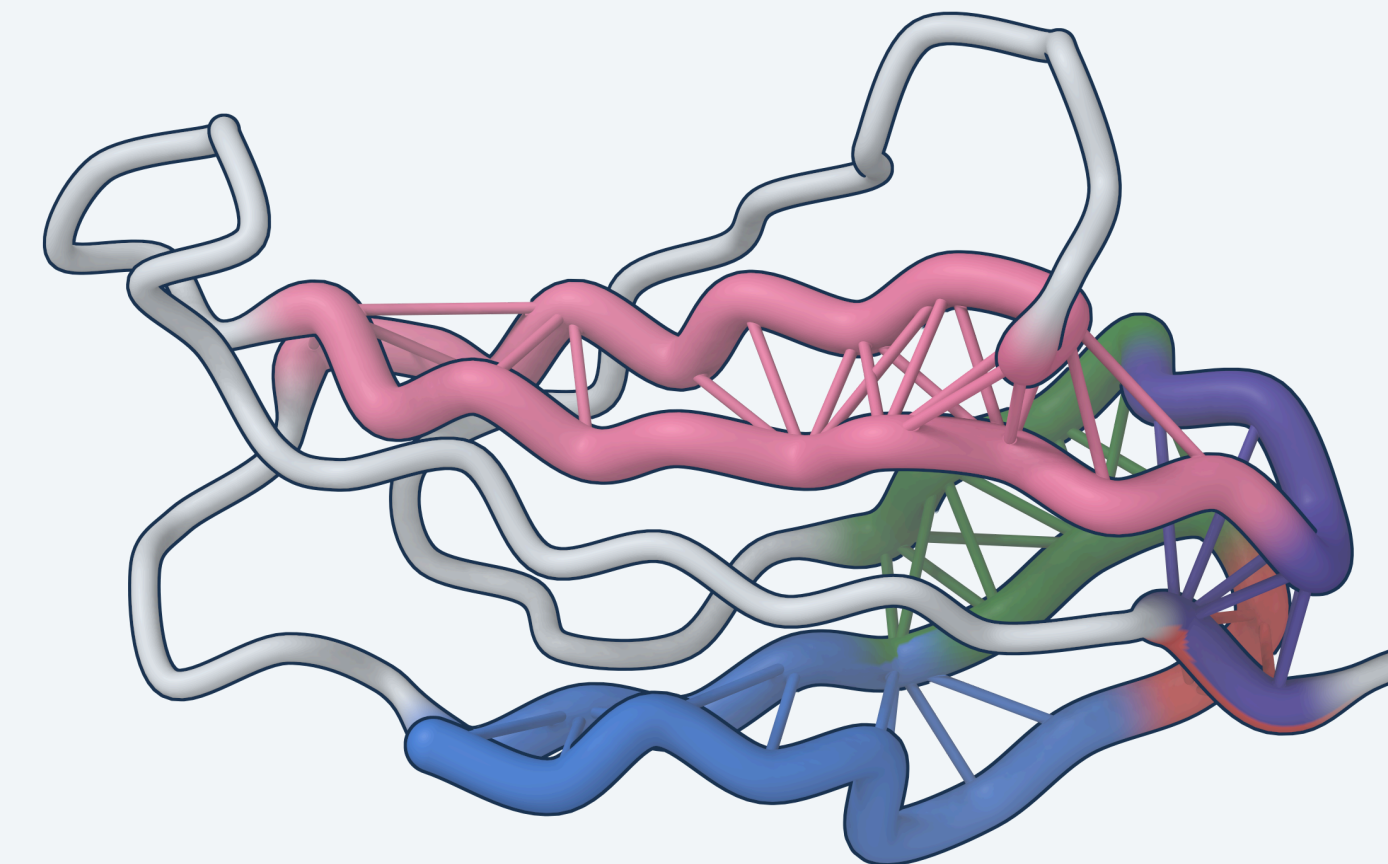
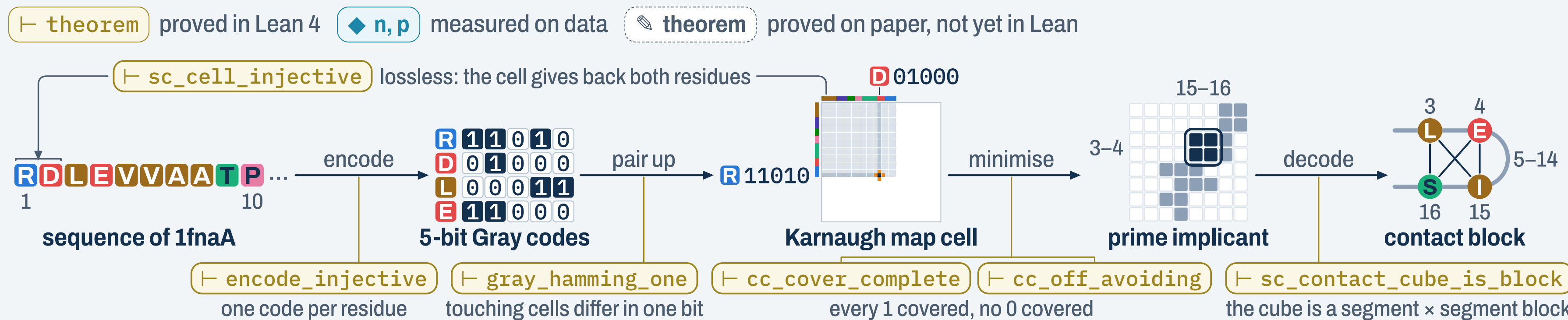
Is KMAP Possible? Formal Verification of Karnaugh Map Encoding for Biological Sequences in Lean 4

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000 The finding

We formalised in Lean 4 the mathematics that turns DNA, RNA and protein sequences into Karnaugh maps: 131 machine-checked theorems, none left unproved. On this verified foundation, exact Boolean minimisation reads a protein's contact map as blocks of touching chain segments, and the same logic uncovers local cooperativity in simulated folding.

Gray-coded grids of sequences, from Petoukhov's genetic Karnaugh matrices¹ to chaos-game maps (FCGR)^{2,3}, are usually taken on trust. Here each step is a checked theorem, and every claim says how it is known.



Strand pairs, found as blocks

- 25-34 with 61-70: 6 blocks
- 1-8 with 14-18: 4 blocks
- 11-14 with 53-56: 3 blocks
- 57-60 with 89-90: 2 blocks
- 9-10 with 89-90: 1 block

Run the proofs on any protein



Pick any structure from the PDB and watch each step run.

001 A code that changes one bit at a time

Binary counting can flip many bits at once: 0111 → 1000 flips four. The reflected Gray code $g(n) = n \oplus (n \gg 1)$ visits every number once and changes exactly one bit per step. DNA needs two bits, and A C G T is itself a Gray cycle.

n	binary	flips	Gray $g(n)$	flips	DNA needs two bits
0	000		000		00 A
1	001	1	001	1	01 C
2	010	2	011	1	10 T
3	011	1	010	1	11 G
4	100	3	110	1	
5	101	1	111	1	
6	110	2	101	1	
7	111	1	100	1	

gray_hamming_one gray_injective gray_cycle_AC...TA

011 A Karnaugh map⁴ is a folded truth table

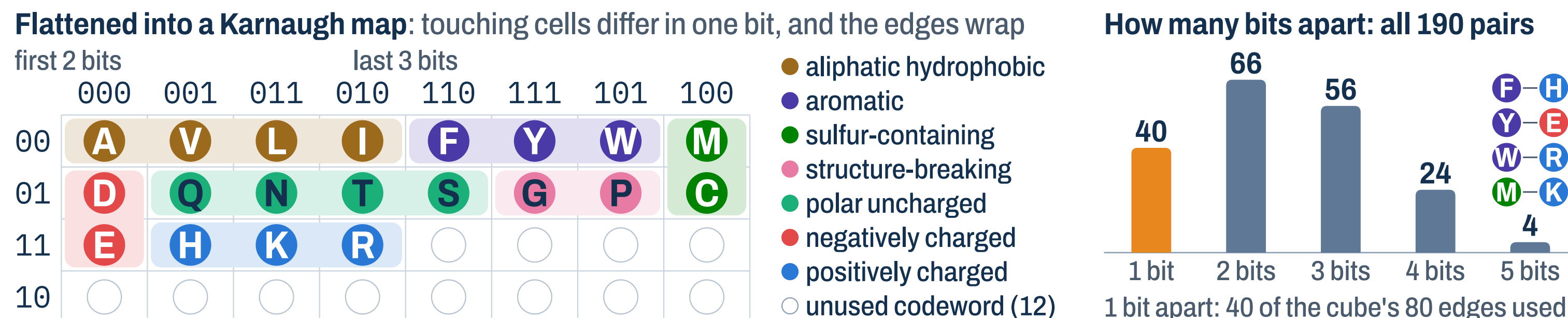
Order a grid's rows and columns by the Gray code and touching cells differ in one variable, even across the edges, which wrap like a torus. Put a k -mer's first letters on the rows and the rest on the columns: each 2-mer owns one cell of a 4×4 map, each 4-mer one of a 16×16 map.

first base	second base	A 00	C 01	G 11	T 10
A 00		AA	AC	AG	AT
C 01		CA	CC	CG	CT
G 11		GA	GC	GG	GT
T 10		TA	TC	TG	TT

dinucEncode_injective tetranucEncode_injective gray_code_preserves_adjacency

010 Amino acids on a five-dimensional cube

Proteins need five bits: 32 codewords for 20 residues. Numbering residues group by group before Gray-coding keeps chemistry close. All 190 pairwise distances are proved: 40 pairs sit one bit apart, only four sit five apart, and the code uses exactly half of the cube's 80 edges.

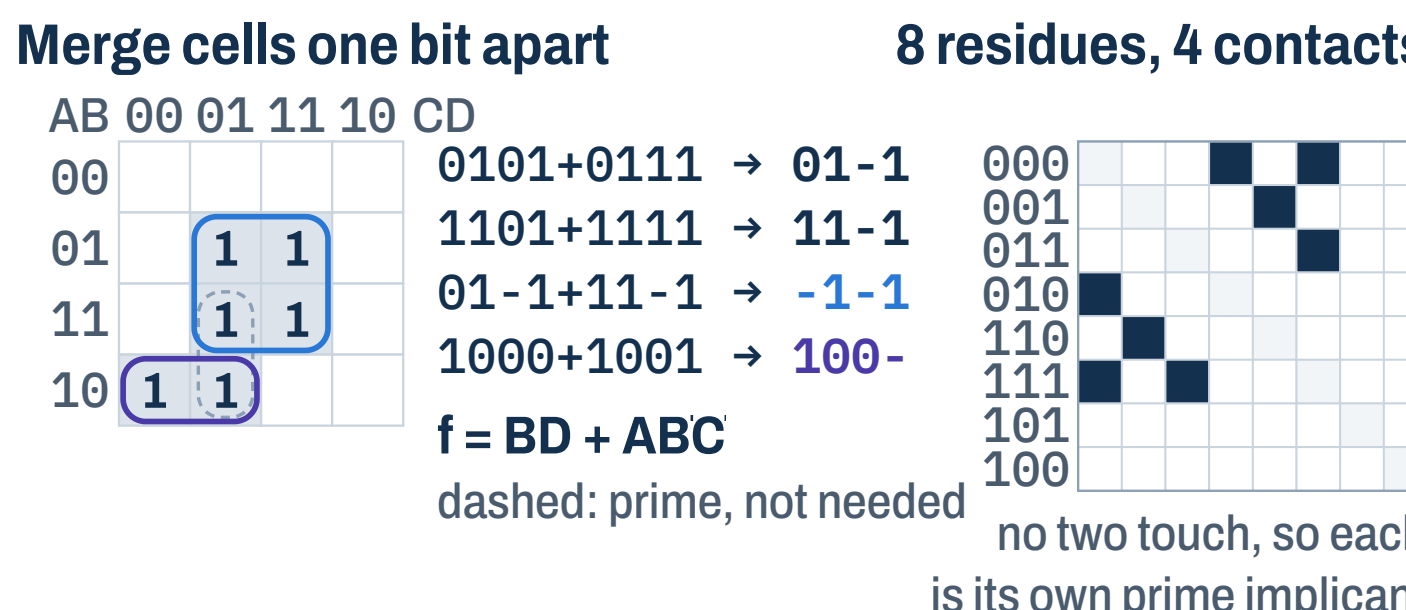


encode_injective unorderedDistanceDistribution max_distance_FH...MK

encoding_edge_coverage

110 Contacts as a Boolean function

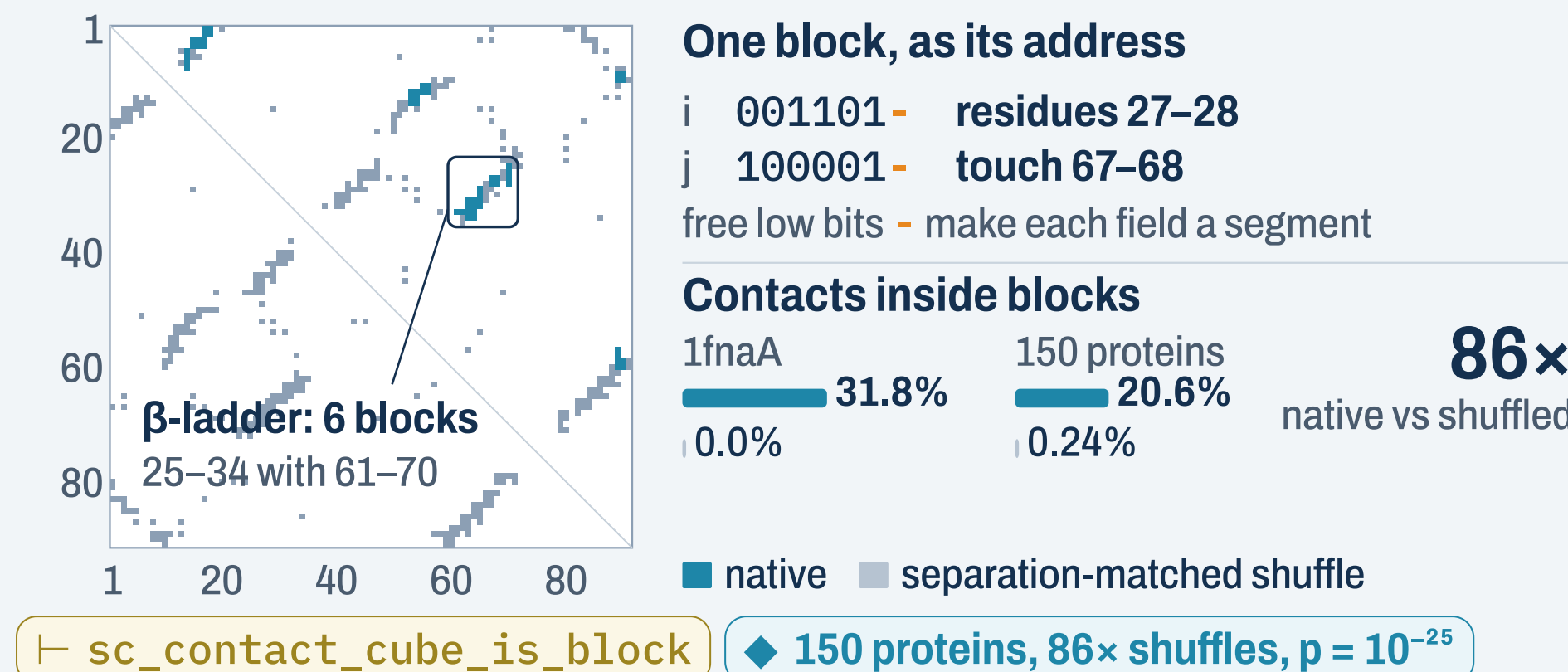
A contact map is a Boolean function: $C(i,j) = 1$ when residues i and j lie within 8 Å. Quine-McCluskey⁵ merges 1-cells one bit apart into ever larger cubes; the cubes that cannot grow, the prime implicants, describe the map most briefly. For an 8-residue example Lean proves the map symmetric, and that no two of its contacts can merge.



contactMap8_symmetric contactMap_no_groupable_minterms cc_cover_complete

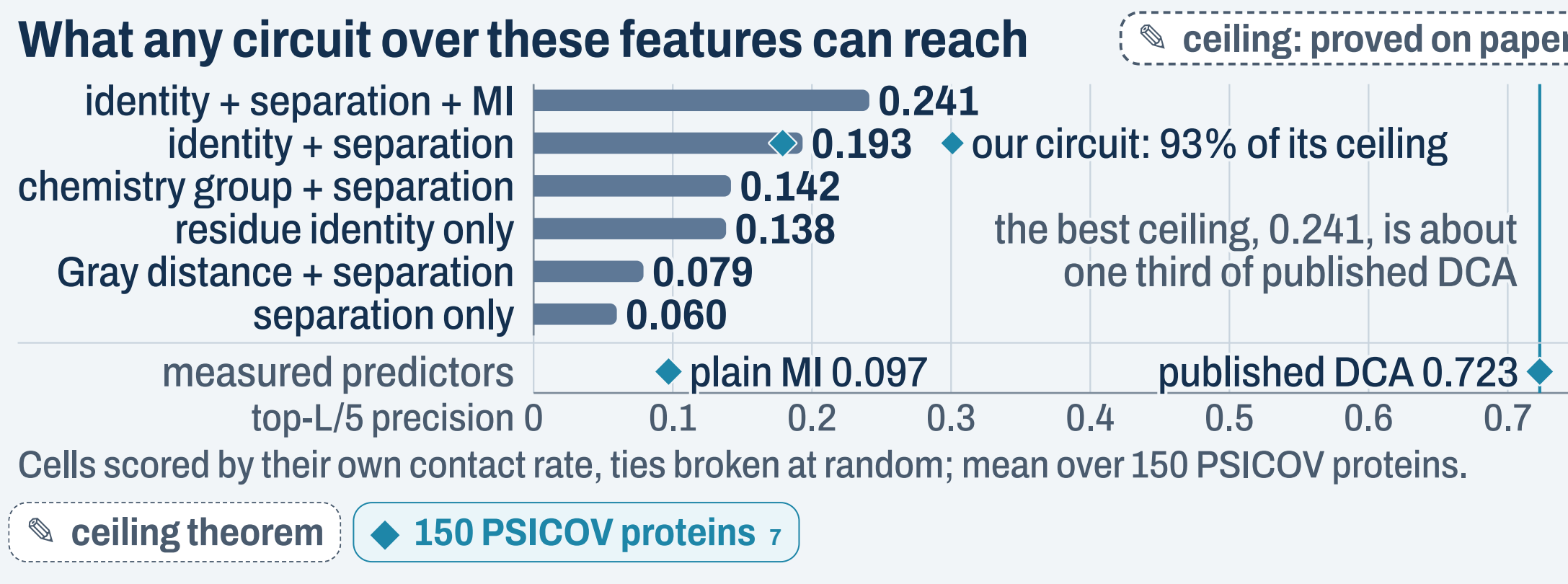
100 What it means for protein structure

With plain position bits, a prime implicant of a contact map is a block: every residue of one chain segment touches every residue of another, the Boolean signature of a β -ladder or a helix packing.



sc_contact_cube_is_block 150 proteins, 86× shuffles, $p = 10^{-25}$

How far residue identity can go, as a bound: a predictor that sees only (residue i , residue j , separation) scores every pair in a cell alike, so cell purity bounds its precision. Coupling features raise the bound.



111 How Lean checks a claim

In Lean 4⁶ a statement is a type and a proof is a program checked by a small kernel. For finite claims, native_decide evaluates every case in compiled code, up to 2^{20} in one theorem, and so also trusts Lean's compiler. No proof uses sorry or adds an axiom of its own.

```
-- Gray code is additive over GF(2)
theorem sc_gray_gf2_linear :
  ∀ x y : Fin 32,
    scGray (x.val ^^^ y.val)
      = scGray x.val ^^^ scGray y.val
:= by native_decide
```

axiom audit: 131 theorems, 0 sorry Python ↔ Lean bridge: 156/156 checks

native_decide standard axioms only no axiom

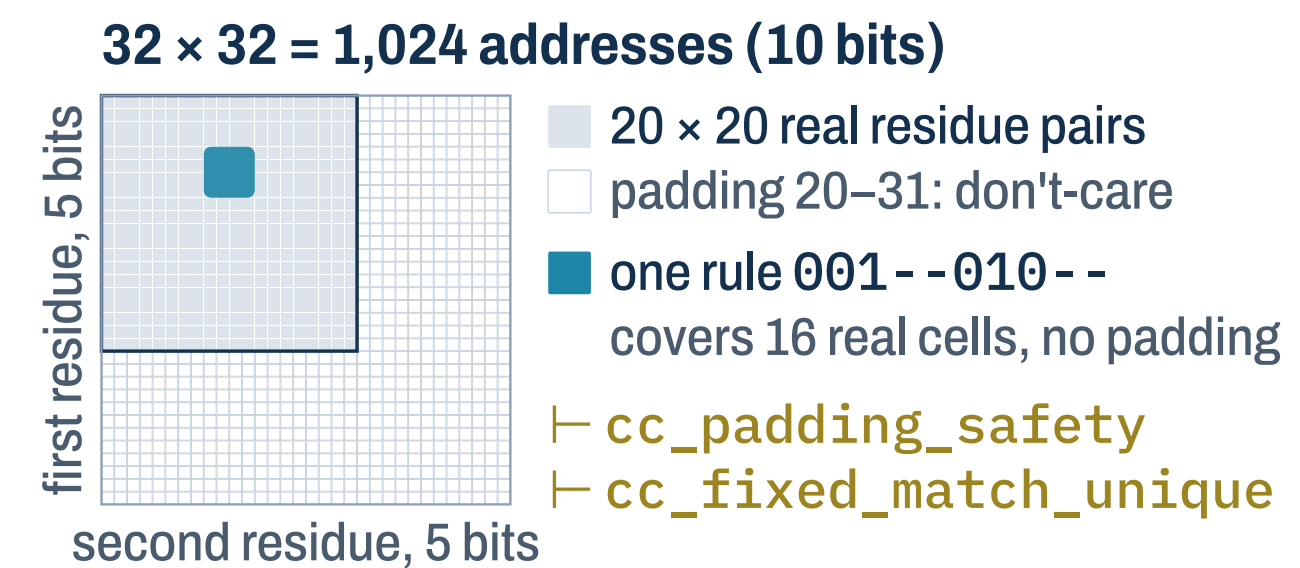
- KmapProofs: 16
- KmerIndexing: 35
- AminoAcidEncoding: 27
- ContactMapCompleteness: 20
- KmapEncodingEquiv: 8
- ContactCircuits: 12
- SequenceCircuits: 13
- 131 theorems
- 100 + 23 + 8

101 What the proofs guarantee

Exact rules on padded maps

Twenty residues need five bits per axis, so a 20×20 residue-pair map is padded to 32×32 with 10-bit addresses. Lean proves that a fully fixed rule matches exactly its own cell and that any rule on real residues fires only inside the 20×20 block: padding never enters a result.

cc_padding_safety cc_fixed_match_unique cc_off_avoiding



Segments hold under either numbering

With its low position bits free, a cube is an unbroken chain segment. Lean proves this for Gray-coded positions as well as plain binary: the two numberings name different segments, never broken ones.

sc_gray_cube_is_also_interval sc_gray_relabels_segments

Same cube, still a segment, different residues

Plain binary cube 10 → residues 8-11

Gray-coded cube 10 → residues 12-15

What the encoding can and cannot change

Since $g(x \oplus y) = g(x) \oplus g(y)$, a Gray-ordered map is a lexicographic (FCGR) map with its cells permuted: every distance-based statistic is identical. The code acts on Hamming distance, where native contacts are one bit apart 1.18× more often than background ($z = 2.5$ against random residue-to-code assignments).

sc_gray_gf2_linear sc_gray_changes_hamming 47,430 contacts, $p = 10^{-77}$

Lexicographic (FCGR) Gray Same 16 counts, moved

A C G T i → g(i) A C T G

CGCGAATTCGCG (1BNA), address order

◆ cophenetic 0.976747 = 0.976747

What Gray changes: pairs one bit apart ($d = 1$)

22.8% native contacts

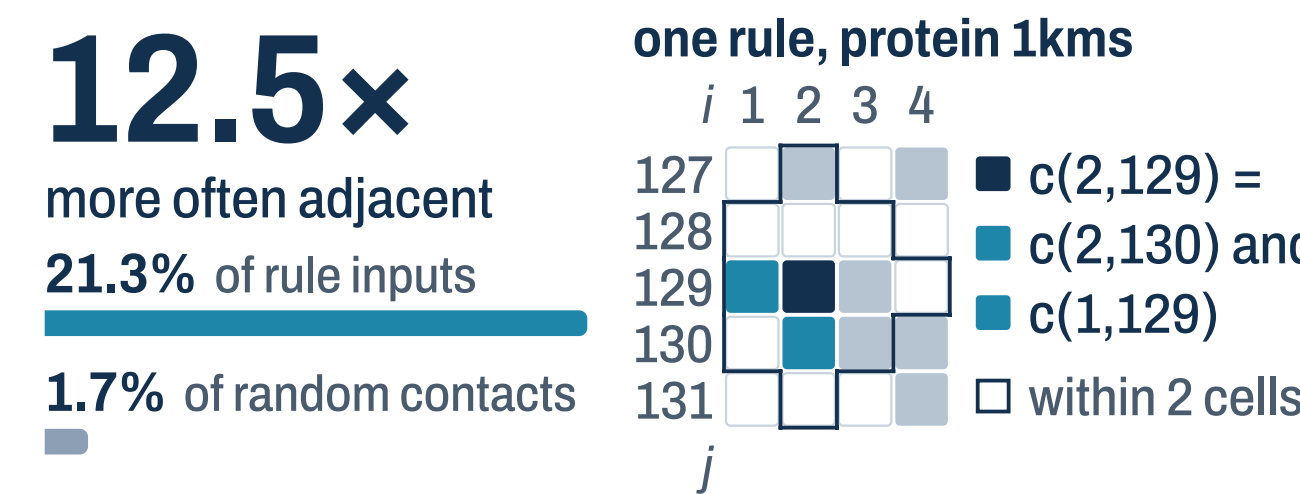
19.4% background pairs

1.18×

The same minimiser on simulations

Run on contact states along structure-based-model⁸ trajectories instead of sequences, Boolean rules link contacts that are neighbours on the contact map: local folding cooperativity, found without being told.

480 proteins, 18,714 rules, 478 above random



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